



United International University
School of Science and Engineering
Department of Computer Science and Engineering
Final Examination Trimester: Fall – 2024
Course: Math -2205 (Probability & Statistics)
Total marks – 40 || Duration – 2 hours

[Note that the number of marks is given in brackets [] at the end of each question or part question. You are requested to answer all the questions in order.]

Q1. (a) A coin is tossed three times. Determine the probability that heads appear on all three tosses. [1]

(b) One digit is randomly selected from 1,2,3,4,5,6,7,8 and 9. Three possible events are defined as [4]

A: the selected digit is a multiple of 3

B: the selected digit is a factor of 8

C: the selected digit is prime number

(i) Find $P(BUC)$

(ii) Is event A and B mutually exclusive? Justify your answer

(c) Given that $P(A) = 0.4$, $P(B) = 0.7$, if the events A and B are independent, then find $P(AUB)$ [2]

(d) A boy walks (W) to school 60% of the time and Cycles (C) 40% of the time. He is late to school(L), [3]
on 5% of the occasions that he walks and he is late on 2% of the occasions that he cycles. Given that
he is late for school, find the probability that he cycles. i.e. find $P(C/L)$

Q2 (a) The following tables show the probability distribution for a random variable X. [3]

x	1	3	9	m
$P(X = x)$	0.4	0.28	0.14	0.18

Given that $E(x) = 5.38$. find the value of m and hence find the variance of X

(b) A typewriter makes, on average, 2 errors per page of typing. In a document of 3 pages, find the probability that he will make at most 2 errors. [2]

(c) A factory makes electronic circuit boards and on average, 4% of them have a minor fault. Find the probability that a random sample of 15 circuit boards has more than 12 circuit boards without any fault. [2]

(d) A petrol station finds that its daily sales, in liters, are normally distributed with a mean 4520 and standard deviation of 560. Find how many days a month (30 days), the daily sales can be expected to exceed 3900 liters. [3]

- Q3 Two sisters make contact using an internet messaging service. The length of time for which they are logged on, in minutes, is modelled by the random variable T with probability density function:

$$f(t) = \begin{cases} \frac{1}{k} (40 - t) & 10 \leq t \leq 30 \\ 0 & \text{otherwise} \end{cases} \quad \text{where } k \text{ is constant.}$$

- (a) Verify that $k = 400$ [2]
- (b) Find the probability that the time they are logged on for is less than 15 minutes. [2]
- (c) Find the corresponding *cdf* (cumulative density function) of the above random variable T. [2]
- (d) Find the mean and median value for time they are logged on for. [4]
- Q4 (a) The label on a certain packet of sweets states the contents are 100 gm. It is known that the standard deviation is 5 gm. The mechanism producing these packets of sweets is checked. From a random sample of ten packets, the mean is 103.8 gm. Find a 99% confidence interval for the mean contents of the packets of sweets. [3]
- (b) A farmer finds that 30% of his sheep are deficient, in particular minerals. He changes their feed and tests 20 sheep to find out if the number has decreased. Carry out a hypothesis test at 10% significance level given that 5 of the sheep are mineral deficient. [3]
- (c) A manufacturer claims that the life of light bulbs produced is normally distributed with mean 800 hours and standard deviation 50 hours. John suspects the claim and purchases 10 light bulbs and found their mean life 780 hours. Carry out a hypothesis test, at 2.5% significance level. [4]

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Related Formulae

Distribution	<i>pmf/pdf</i>
Binomial	$f(x) = n_c p^x (1 - p)^{n-x}; x = 0, 1, 2, \dots, n$
Poisson	$f(x) = \frac{\lambda^x e^{-\lambda}}{x!}; x = 0, 1, 2, \dots$
Uniform	$f(x) = \frac{1}{b-a}; a \leq x \leq b$
Normal	$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}; -\infty < x < \infty$
Confidence interval for	(i) Population mean with known variance $\bar{x} - z \frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + z \frac{\sigma}{\sqrt{n}}$ (ii) Population portion $\bar{p} - z \sqrt{\frac{\bar{p}(1-\bar{p})}{n}} < p < \bar{p} + z \sqrt{\frac{\bar{p}(1-\bar{p})}{n}}$